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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 9

Computer Memory

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CHAPTER 9

Computer Memory

Computer Awareness

IN THIS CHAPTER

- Memory cells and addresses
- Memory hierarchy
- Bits, bytes and words
- Character coding
- CPU registers
- Cache memory

Cold Boot = Power ON

Warm Boot = Restart

POST = Hardware Check

Linux = Open Source

A computer is an electronic device that accepts raw data (input), processes it according to a set of instructions, and produces meaningful information (output).

In simple words, a computer takes data from the user, stores it temporarily in memory, processes it using the CPU, and then provides the desired result.

WORKING OF A COMPUTER

INPUT → PROCESSING → OUTPUT

During processing, the computer stores both data and instructions in memory.

Example

Input: Student marks entered into a computer

Processing: Calculation of total marks and percentage

Output: Result displayed on the screen

Important Exam Point

A computer is also known as a Data Processing System because it converts raw data into useful information.

WHAT IS MEMORY?

Memory is the storage area of a computer where data, instructions, and information are stored. It performs a function similar to the human brain by storing information and providing it when required.

Functions of Memory

- Stores data
- Stores instructions
- Stores intermediate results
- Stores final output

MEMORY CELLS

Computer memory is divided into many small storage locations called Memory Cells.

The memory unit of the CPU consists of a large number of cells called locations

Each memory cell:

- Stores data or instructions
- Has a unique address
- Is identified by its address

MEMORY ADDRESS

A memory address is a unique number assigned to each memory location.

Addresses usually start from 0 and continue up to the memory size minus one.

Example

Address	Data
0	A
1	B
2	C

CLASSIFICATION OF MEMORY

Computer Memory

- Register Memory
- Cache Memory
- Primary Memory
- RAM
- ROM
- Secondary Memory

Memory Hierarchy

Fastest ↓ Register ↓ Cache ↓ RAM ↓ Secondary Storage ↓ Slowest

Register → Cache → RAM → Storage

(Fastest to Slowest)

DATA REPRESENTATION IN MEMORY

Computers understand only Binary Language.

Binary language consists of two digits:

- 0 = OFF State
- 1 = ON State

All data, instructions, images, videos, and programs are stored as combinations of 0s and 1s.

Write Operation

The process of storing data into memory is called Writing.

Example: Saving a file.

Read Operation

The process of retrieving data from memory is called Reading.

Example: Opening a saved file.

WORD LENGTH

Word length is the number of bits a processor handles as a natural unit. A memory address may refer to a byte or a word, depending on the architecture.

Common word lengths are:

- 8-bit
- 16-bit
- 32-bit
- 64-bit

Important Exam Point

Larger word length means the CPU can process more data at one time.

BASIC MEMORY UNITS

•Bit

- Bit stands for Binary Digit.

- It is the smallest unit of memory.

- A bit can store either: 0 or 1

•Nibble

- 1 Nibble = 4 Bits

- Example:

- 1010 = 1 Nibble

•Byte

- 1 Byte = 8 Bits

- A byte can represent 256 different combinations.

- Important Correction

- ASCII uses 7 bits for character coding, but a Byte always consists of 8 bits.

- Many students confuse these two concepts.

•Word

- A Word is a group of bits processed together by the CPU.

CHARACTER CODING SYSTEMS

- ASCII

- ASCII stands for:American Standard Code for Information Interchange

-

- Uses 7 bits

-

- ASCII is a historically important character code; Unicode is widely used in modern systems.
- EBCDIC

- EBCDIC stands for:Extended Binary Coded Decimal Interchange Code

-

- Uses 8 bits

-

- Developed by IBM

-

- Mainly used in mainframe computers

REGISTER MEMORY

Register Memory is the smallest and fastest memory in a computer.

It is located inside the CPU and is not considered part of the main memory.

Registers temporarily store data, instructions, and addresses required by the CPU.

Features of Register Memory

Feature	Description
Location	Inside CPU
Speed	Fastest
Capacity	Very Small
Cost	Highest
Access Time	Lowest

Important Exam Point

Registers hold the operands and intermediate results used in many CPU operations; the exact data path depends on the architecture.

FLIP-FLOP

Registers are made up of electronic circuits called Flip-Flops.

1 Flip-Flop stores 1 bit of data.

Example

8 Flip-Flops = 8 Bits = 1 Byte

TYPES OF REGISTERS

1. Data Register (DR)

In the illustrative 16-bit model, the data register is 16 bits wide. Register widths vary between processors.

Stores data that are to be processed.

Functions:

- Holds input data
- Holds output data
- Exchanges data with peripheral devices(all inputs and output devices)

Peripheral Devices: Keyboard, Mouse, Scanner, Printer, Monitor, etc.

2. Program Counter (PC)

Stores the address of the next instruction to be executed.

Function:

Maintains the sequence of program execution.

3. Instruction Register (IR)

In the illustrative 16-bit model, the instruction register is 16 bits wide. Instruction-register widths vary between processors.

Stores the instruction currently being executed.

Functions:

- Receives instruction from memory
- Sends instruction to Control Unit
- Helps in decoding and execution

4. Accumulator Register (AC)

Stores intermediate and final results generated by the CPU.

Example:

10 + 20 = 30

The result (30) is temporarily stored in the Accumulator.

5. Address Register (AR)

Stores addresses of memory locations where data or instructions are located. 6. I/O Address Register

Stores the address of an input or output device.

7. I/O Buffer Register

Transfers data between CPU and input/output devices.

COMPUTER BUSES

Registers communicate using communication pathways called Buses.

A Bus is a set of wires that transfers data, addresses, and control signals.

TYPES OF BUSES

1. Address Bus

Used to transfer memory addresses.

Address Bus is generally UNIDIRECTIONAL.

CPU → Memory

2. Data Bus

Used to transfer data.

Data Bus is BIDIRECTIONAL.

CPU ↔ Memory

3. Control Bus

Used to transfer control signals.

Examples:

- Read Signal
- Write Signal
- Interrupt Signal

CACHE MEMORY

Cache Memory is a high-speed memory located between the CPU and RAM.

It is:

- Faster than RAM
- Slower than Registers
- Smaller in size

Function:

Stores frequently used data and instructions.

This reduces CPU waiting time and increases system performance.

Important Exam Point:

Cache acts as a BUFFER between CPU and RAM.

WORKING OF CACHE MEMORY



Cache supplies a fast intermediate level between the CPU and main memory.

Without Cache

CPU → RAM → Data

With Cache

CPU → Cache → Data

If data is available in Cache Memory, the CPU accesses it immediately without searching RAM.This process significantly improves performance.

TYPES OF CACHE MEMORY

1.L1 Cache (Level 1)

- Fastest Cache
- Smallest Size

- Located inside CPU

2.L2 Cache (Level 2)

- Larger than L1
- Slightly slower

3.L3 Cache (Level 3)

- Largest Cache
- Shared among processor cores
- Slower than L1 and L2

QUICK REVISION NOTES

- Computer is a Data Processing System.
- Memory stores data and instructions.
- Memory consists of cells.
- Each cell has a unique address.
- Bit is the smallest memory unit.
- 1 Nibble = 4 Bits.
- 1 Byte = 8 Bits.
- ASCII uses 7 bits.
- EBCDIC uses 8 bits.
- Register is the fastest memory.
- Cache acts as a buffer between CPU and RAM.
- Program Counter stores the next instruction address.
- Accumulator stores processing results.
- Address Bus is unidirectional.
- Data Bus is bidirectional.

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

1.Fastest Memory → Register

2.Second Fastest Memory → Cache

3.Smallest Memory Unit → Bit

4.1 Byte = 8 Bits

5.ASCII → 7 Bits

6.EBCDIC → 8 Bits

7.Program Counter stores next instruction address

8.Accumulator stores results

9.Cache stores frequently used data

10.Flip-Flop stores 1 bit

PRIMARY MEMORY (MAIN MEMORY)

Primary Memory is the memory directly accessible by the CPU. It is also known as:

- Main Memory
- Internal Memory
- Semiconductor Memory

Primary memory stores data and instructions that are currently being used by the CPU.